

Ronald Eduardo Barberi Riaño

Data Scientist & Data Engineer | Machine Learning · Big Data/ETL · Analytics

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Professional Summary

Senior Data Scientist and Data Engineer with 6+ years of experience across data analytics, data science, and data engineering in banking, payments, technology, and BPO. End-to-end data lifecycle: batch and streaming ingestion (PySpark, Apache NiFi, AWS Glue) over 5+ billion records, terabyte-scale feature engineering, inferential statistical analysis, production machine learning models (churn, NBO/NBA, classification) with MLOps on AWS SageMaker, and KPI reporting for senior leadership. Drove 275% campaign growth, an estimated 20% lift in promotion effectiveness, and an estimated 8% recovery of churning customers.

Technical Skills

Analytics & BI: SQL (SQL Server, MySQL, PostgreSQL, Athena/Presto, Hive, Impala), Power BI, Tableau, Looker, descriptive & inferential statistics, exploratory data analysis (EDA), R, KPI reporting, advanced Excel (macros), Matplotlib, Seaborn, Erlang C

Data Engineering: Python, PySpark / Apache Spark, Pandas, Polars, NumPy, batch & streaming ETL/ELT, Apache Airflow, dbt, Apache Kafka, Apache NiFi (Groovy), Databricks, Snowflake, BigQuery, Cloudera (Hive, Impala, HUE), AWS Glue, Amazon EMR, Amazon S3, APIs, web scraping (Selenium), Java

Data Science & ML: scikit-learn, XGBoost, LightGBM, TensorFlow / Keras, classification models, recommender systems (NBO/NBA), churn modeling, feature engineering, MLflow, model monitoring (MLOps), Amazon SageMaker (JumpStart), computer vision (OpenCV, MediaPipe), LLM agents (Claude)

Cloud, DevOps & APIs: AWS (S3, EC2, EMR, Glue, Athena, SageMaker, CloudWatch, CodeCommit), Cloudera, Azure (coursework), Docker, Git / GitHub, GitHub Actions (CI/CD), FastAPI, Django, Linux (server administration), Automation Anywhere 360 (RPA)

Practices: Scrum, clean code, PEP 8

Experience

Business Data Coordinator (Bilingual)

07/2026 – 10/2026

Samsung

- Built binary classification models in Python and PySpark to select targeted promotions for each customer, driving an estimated 20% lift in promotion effectiveness.
- Created AI skills and agents with Claude (LLMs) that automated ~15 monitoring, migration, and operations processes, saving ~10 hours of work per day.
- Built batch and streaming ETL pipelines in PySpark processing ~15 GB per flow, storing the processed data in MySQL for business consumption.
- Designed the KPI reports that the company's President's Office reviews in detail to track business performance.
- Automated processes with Python and Automation Anywhere 360, and configured the Linux servers and MySQL databases that run them.

Tech stack: Python, PySpark, MySQL, Linux, Automation Anywhere 360, Claude (LLM)

Data Scientist (Mid-level)

11/2025 – 07/2026

Itaú (via Multiplica)

- Built a feature engineering framework in Python and Spark on Amazon EMR that processes terabytes of data and produces the ML master tables for ~7 projects, saving ~6 hours of data preparation.
- Maintained and ran production models on AWS SageMaker: customer promotion recommenders (Next Best Offer and Next Best Action), customer churn (Churn 360), and SPOT.
- Implemented continuous monitoring of production models following MLOps standards with SageMaker and CloudWatch, ensuring quality and security and issuing retraining recommendations.
- Operated and maintained high-volume ETL pipelines with AWS Glue, S3, and Athena (SQL/Presto) supporting reporting and operational processes.
- Developed a high-level Python transformation framework for univariate statistical analysis and data treatment, reusable across modeling projects.

Tech stack: AWS (SageMaker, JumpStart, EMR, Glue, Athena, S3, EC2, CloudWatch, CodeCommit), Python, Spark

Data Engineer

12/2024 – 11/2025

Credibanco (via Nexos)

- Designed and built ETL pipelines with PySpark and Apache NiFi on Cloudera (Hive, Impala) to transform and transfer 5+ billion records (~150 columns) across multiple projects.
- Identified the root cause of merchant customer churn with inferential statistical analysis in Python; the proposed corrective actions recovered and retained an estimated 8% of customers.
- Refactored the Hub Connect APIs with new consumption logic and business rules, increasing capacity and freeing Cloudera resources (estimated savings of COP 9 million per month).
- Implemented hashing of sensitive values and streaming data transfer with Apache NiFi, protecting sensitive data across pipelines.

Tech stack: Cloudera (Hive, Impala, HUE, CDSW, Zeppelin), PySpark, Apache NiFi, AWS, Java, Python (scikit-learn, Pandas)

Data Scientist / Data Mining

04/2022 – 12/2024

Customer Operation Success (COS)

- Led a Scrum team on database cleansing for the Claro, Laika, and portfolio (cartera) campaigns, driving 275% growth and making them the most profitable campaigns of the period.
- Developed Python and PySpark ETL pipelines extracting data from APIs, web scraping (Selenium), and servers, with streaming processing.
- Built EDAs and predictive models (scikit-learn, TensorFlow) to size and profile databases.
- Created the business KPI reporting with standardized templates, and automated processes running on Linux servers (virtual machines).

Tech stack: SQL, Python, PySpark, Pandas, NumPy, Selenium, scikit-learn, TensorFlow, Matplotlib, Seaborn, Erlang C, Git/GitHub, Docker, Linux

Prior Experience

- **Data Analyst** – Colombian National Army (09/2020 – 02/2022): SQL Server stored procedures, views, and validation queries; Excel reports with macros to cut turnaround time.

Featured Projects

- **Multi-source ETL Pipelines** – github.com/RonaldBarberi/structured_data_etls
Python and SQL ETL pipelines extracting data from websites (Selenium), MySQL databases (SQLAlchemy), and local files, with stored procedures for database cleansing.
- **Machine Learning & Computer Vision** – github.com/RonaldBarberi/data_science
Imbalanced credit card usage classifier (Random Forest, 0.74 cross-validated ROC-AUC), grocery cost regression validated on unseen periods ($R^2 = 0.84$), and hand-gesture recognition with OpenCV and MediaPipe (98.7% accuracy), built with scikit-learn pipelines.
- **Exploratory Analysis & Statistical Validation** – github.com/RonaldBarberi/data_analytics
EDA on banking and BPO-sector data with cleaning and statistical validation, delivering model-ready datasets for binary classification.

Education

Statistics

01/2025 – 12/2027 (expected)

Fundación Universitaria Los Libertadores – Bachelor's degree (in progress, 6th semester)

Specialization in Applied Statistics as the degree completion option. Expected graduation: H2 2027.

Computer Engineering

Escuela Superior de Empresa, Ingeniería y Tecnología (ESEIT) – 4 semesters completed

Focus: software infrastructure, servers, databases, and Big Data. Transferred to Statistics.

Certifications

- Azure Data Engineer Technologies – Udemy (12/2025)
- Big Data & Spark: Data Engineering with Python and PySpark – Udemy (12/2025)
- AWS Solutions Architect (AWS Certified Solutions Architect exam prep) – Udemy (12/2025)

Languages

Spanish: Native | English: Intermediate (B1)